Decision Makers' Connect

SOLAR ROADWAYS: Feasible For India?

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Solar roads could put an end to India's electricity woes. But, to make these technically and financially viable, there is a long list of challenges to overcome—ranging from indigenous solar system development to infrastructure and investment

"People don't want to do that (build houses or offices) near a nuclear power plant. So, there's quite a big keep-out zone, and when you factor the keep-out zone into account, the solar panels put on that area would typically generate more power than that nuclear power plant."

—Elon Musk

India is moving full throttle on solar projects. This has set the stage for higher domestic manufacturing and thus lower cost of solar products. With consistent availability of sunlight throughout the year (for about 300 days), one of the most unique proposals is the implementation of solar surfaces for roads in India. This has the potential to generate massive megawatts of energy that can power numerous buildings and other infrastructure. However, developing solar panel roads is not easy and requires more clarity and planning on many aspects before implementation.

The benefits

A few years ago, ideas were invited to install rooftop solar panels on the roads of Gujarat to generate electricity from solar energy. One proposal came from the Gujarat Energy Research and Management Institute (GERMI) to install solar panels across 205km of the Ahmedabad-Rajkot highway to generate 104 megawatts of electricity. The energy generated could be used to power the highway infrastructure, light the roads and also run a bit of the adjoining buildings. The project is in planning stage.

Roads with solar surfaces can immensely transform the Indian power situation. Bhargav Vyas, vice president, PV Power Tech, says, "Implementation of solar roadways should start from the highways in India. It will be easier to receive permission and work quickly in those areas. As the exposure to sunlight will be much stronger than in closed areas, energy generation will be substantial."

India faces the problem of electricity shortage even today. Power is yet to be introduced in some regions. Certain reports suggest around 240 million Indians still live in the dark, ranking India among countries with one of the highest unelectrified populations in the world. Implementation of projects like solar roads can substantially improve the situation. While solar street lights and solar rooftops are being deployed rapidly, extensive solar surface on roads can become the central point for powering a larger proportion of national infrastructure.

In the long run, such projects can massively reduce the use of grid energy, cut down fossil fuel consumption exponentially and bring down the national carbon footprint dramatically.

If developed with good-quality resilient material, solar panel roads can eliminate the frequent damage-and-maintenance that is associated with pavement roads. This would save a lot of administrative fund, adding to the payback period on the investment. Issues of potholes,

Solar bike stretch in the Netherlands